Test card: flat wings and the swift-like outline

MODEL = from our computer model, not a fact. FIRM = follows from symmetry or from the project's own tests. Whether to bend the trailing edge up, and what drop height to use, are Kai's decisions; this card recommends neither.

1. Specimens: CG table, mm aft of the root leading edge

F45 (printed, span 250 mm) CG 36 to 49 mm, 1 mm steps (14 settings)

AVL neutral point with the tip fins: 43.9 mm (outline only, a place to start, not a result)

F60 (printed, span 250 mm) CG 34 to 44 mm, 1 mm steps (11 settings)

AVL neutral point with the tip fins: 39.6 mm (outline only, a place to start, not a result)

B45 (foamboard, span 400 mm) CG 57 to 78 mm, 1 mm steps (22 settings)

AVL neutral point with the tip fins: 70.2 mm (outline only, a place to start, not a result)

Stop rule (unchanged): after the table, keep going aft in 1 mm steps while any setting still glides or still dives. Stop when, after at least one gliding setting, two settings in a row give no steady glide, or at the bare-aircraft balance point. Releases: 5 per setting, 10 at any setting that glides. Weigh the glider at each setting.

2. Swift-like outline (bird, version 2)

Tail bend 5 deg, trailing edge UP. Trim CG at lift coefficient 0.3: 26.6 mm (printed, span 250 mm) and 42.6 mm (foamboard, span 400 mm) aft of the root leading edge. MODEL (AVL outline model): a reference point, not a CG window, not a prediction.

3. Order of tests

a. Weigh every part. Find the CG on a knife edge to 0.5 mm. Check the wing is flat and not twisted.
b. LEVEL HAND GLIDE first: gross trim only. One or two seconds sort a dive from a pitch-up; they cannot show a repeatable glide.
c. NOSE-DOWN DROP from height: hold it nose straight down, let go at rest. Same height every time. Height is Kai's decision.
d. Later, quad release (not built yet): hang level, release in forward flight, not from a hover (FLAT_TEST.md procedure).

Outcome words, exactly: steady glide, dive, pitch-up or stall, pitch oscillation, tumble, roll-off left, roll-off right, yaw wander.

What you see -> what it means -> what to change next

MODEL = from our computer model, not a fact (results/runs/rigid_glider/pullout/). FIRM = follows from symmetry or from the project's own tests. Bending the trailing edge up and the drop height are Kai's decisions.

What you see	What it probably means	What to change next
Darts straight down (log: dive)	FIRM: released exactly nose-down with the CG ahead of the lift point, a flat symmetric wing has no moment to start pitching, so it stays a dart. Says nothing against the wing.	Do the level hand glide first. Then move the CG aft 1 mm at a time (kit table). Bending the trailing edge up is Kai's decision.
Swoops, then glides (log: steady glide)	MODEL: a first swoop of about 6 m or more is expected even for a wing that will glide, so a short drop cannot tell 'never pulls out' from 'needs more height'.	Log distance and time; repeat 5 times (10 if it glides). Drop height is Kai's decision.
Mushes: flat, slow, steep descent (log: steady glide, note 'mush')	Flying near stall. The model cannot say at which CG an untabbed flat wing settles. Cause not established.	Check flatness and twist. Try the neighbouring settings, 1 mm forward and aft, and compare.
Nose up, stalls, repeats (pitch-up or stall; pitch oscillation)	Seen behind the neutral point. On a swept, tapered wing the tips may let go first (unknown). Pitch oscillation is about the balance point and the wing, not the fins.	Check twist. Move the CG forward 1 mm. Do not blame the fins.
Tumbles (tumble)	In the published plate study, tumbling came with the centre of mass furthest aft (unswept plates; not our wings).	Move the CG forward. The stop rule decides when the sweep ends.
Turns or spirals (roll-off left / right; yaw wander)	Yaw wander (nose swings side to side) points at the fins. Roll-off points at a twisted or uneven wing.	Check the fins are square, the wing flat and not twisted. Re-fly.

Flight log

Date _____ Place _____ Wind _____

First measured masses (g): F45 _____ F60 _____ B45 _____ bird _____

Which wing was the 'about 15 g' is still unconfirmed: write the weighed masses here.

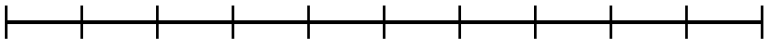
Launch: H = level hand throw, N = nose-down drop. CG = mm aft of the root leading edge (or rail tick). Tab / bend in degrees, trailing edge up.

Result (write the letter): S = steady glide, D = dive, P = pitch-up or stall, O = pitch oscillation, T = tumble, L = roll-off left, R = roll-off right, Y = yaw wander. A mush is logged as S with a note in Notes.

No	Wing	Mass g	CG mm	Tab / bend deg	Launch H/N	Height m	Result	Dist m	Time s	Notes
1										
2										
3										
4										
5										
6										
7										
8										
9										
10										
11										
12										
13										
14										
15										
16										
17										
18										
19										
20										

Angle gauge, 1:1

Print at 100 % and check the 100 mm bar below. Cut out each wedge along its three sides. Lay one long edge along the wing and the other along the bent trailing-edge tab or the bird tail: the gap should close at the tip. The angle is the angle between the two long edges (computed, not drawn by eye).



Scale bar: this line must measure exactly 100 mm. If it does not, print at 100 % (actual size).

2 deg

long edges 70 mm, end gap 2.4 mm

2.5 deg

long edges 70 mm, end gap 3.1 mm

4 deg

long edges 70 mm, end gap 4.9 mm

5 deg

long edges 70 mm, end gap 6.1 mm

6 deg

long edges 70 mm, end gap 7.3 mm

7.5 deg

long edges 70 mm, end gap 9.2 mm

8 deg

long edges 70 mm, end gap 9.8 mm